Implementation of Forward Chaining

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RULES = [  # (premises, conclusion)
    (["frog"], "green"),
    (["croaks", "eats_flies"], "frog"),
    (["chirps", "sings"], "canary"),
    (["canary"], "yellow"),
]
FACTS = {"croaks", "eats_flies"}


def forward_chain(facts, rules, goal=None):
    facts = set(facts)
    changed = True
    while changed:
        changed = False
        for premises, concl in rules:
            if concl not in facts and all(p in facts for p in premises):
                print(f"  Rule {premises} -> {concl}  fires, new fact: {concl}")
                facts.add(concl)
                changed = True
                if goal and concl == goal:
                    return facts
    return facts


def program9():
    print("\n=== 9. FORWARD CHAINING ===")
    print("Facts:", FACTS)
    result = forward_chain(FACTS, RULES)
    print("All derived facts:", result)
    print("Is 'green' derived? ->", "green" in result)
    print("Is 'yellow' derived? ->", "yellow" in result)


if __name__ == "__main__":
    program9()
```